The Electricity Journal 37 (2024) 107373

Contents lists available at ScienceDirect

The Electricity Journal

journal homepage: www.elsevier.com/locate/tej

House and socio-demographic features vs. electricity consumption time series in main heating mode classification

Hannu Huuki [a,b,*], Enni Ruokamo [a,b], Maria Kopsakangas-Savolainen [a,b], Nadezda Belonogova [c], Araavind Sridhar [c], Samuli Honkapuro [c]

[a] Finnish Environment Institute, Oulu Office, P.O.Box 413, FI-90014 University of Oulu, Finland
[b] Department of Economics, Accounting and Finance, University of Oulu Business school, P.O.Box 4600, FI-90014 University of Oulu, Finland
[c] LUT University, P.O.Box 20, FI-53851, Lappeenranta, Finland

ARTICLE INFO

Keywords:
Home heating system
Time-series classification
House and socio-demographic features

ABSTRACT

Demand-side flexibility is crucial for integrating variable renewable energy sources cost-effectively. Home heating systems determine the potential for flexibility in individual households. We examine different approaches to classify heating systems in Finland and find that using hourly electricity consumption time series is more effective than house and socio-demographic features. Classification based on electricity consumption data achieves higher precision (0.62) and recall (0.64) than house and socio-demographic features (0.41 and 0.43, respectively). Therefore, the availability of electricity consumption time series data should be considered from a competition policy perspective due to its value in estimating flexibility potential.

1. Introduction

The share of variable renewable energy is increasing rapidly globally (IEA, 2022 b). In Finland, 8.1 TWh of electricity (9.4% of consumption) was generated by wind power in 2021 (Finnish Energy, 2022), and the expected wind power generation varies between 51 - 132 TWh/a in the Finnish TSO's future power system scenarios for 2035 (Fingrid, 2022). Intermittent renewable energy resources challenge the efficient operation of the electricity system if grid balancing is based purely on the supply side resources (Huuki et al., 2020). For cost-efficient operation of the energy system, it is crucial that potential balancing resources from the demand-side are activated as well. Consequently, a need exists for advanced methods to screen potential demand-side flexibility sources.

Globally, heating (homes, industry and other) accounted for around 50% of energy demand in 2021 (IEA, 2022a). Due to the geographical location in up north, heating of spaces and domestic water accounts for the largest share (82% in 2021) of energy demand in the household sector in Finland (Statistics Finland, 2022). On the other hand, Finnish buildings are well constructed and properly insulated. As a result, the building stock offers a good source for demand-side flexibility. Currently there are more than 570,000 electrically heated buildings in Finland (Känsälä and Hammar, 2018). The importance of electricity for heating will continue to be emphasized in the future, as heat pump sales are expected to increase in net zero pathways (Rosenow et al., 2022).

Previous literature has examined in detail household preferences for different heating systems and the determinants for system adoption in different countries (for a literature review see Ortega-Izquierdo et al. (2019)). For example, heating system preferences and adoption choices have been studied in Germany by Michelsen and Madlener (2012), in Ireland by Meles et al. (2022), in Sweden by Mahapatra and Gustavsson (2008), and in Finland by Ruokamo (2016), Räihä and Ruokamo (2021) and Sahari (2019). This literature shows that heating system choices are linked with several socio-demographic and home features, such as income (Michelsen and Madlener, 2012; Ruokamo, 2016), education (Sahari, 2019), house location (Ruokamo, 2016; Räihä and Ruokamo, 2021), house floor area and house age (Michelsen and Madlener, 2012). However, a research gap exists in quantifying the performance of households' heating mode classification based on socio-demographic background and house characteristics information.

Detailed consumption data is needed for the analysis of household's flexibility potential. In some countries, such as Finland, smart meters are already installed in all consumption locations (ACER, 2022). Smart meters provide data that enables analysis of consumption profiles of different consumption units. Time series data from smart meters is utilized in several consumption pattern identification techniques. Tureczek and Nielsen (2017) find in their literature review that most studies use

---

* Corresponding author at: Finnish Environment Institute, Oulu Office, P.O.Box 413, FI-90014 University of Oulu, Finland.
E-mail address: hannu.huuki@syke.fi (H. Huuki).

https://doi.org/10.1016/j.tej.2024.107373
Received 25 August 2023; Received in revised form 29 January 2024; Accepted 30 January 2024
Available online 8 February 2024
1040-6190/© 2024 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

unsupervised learning methods in customer group identification. A common method is K-means clustering and its variations. Notably, studies using supervised classification are also using unsupervised clustering for starting conditions (Tureczek and Nielsen, 2017).

Prior work using supervised learning methods in household classification based on electricity consumption data is still narrow. Carrol et al. (2018) employ smart meter data for the household composition classification task. Their study indicates that single-person households can be identified based on electricity consumption patterns. Wang et al. (2019) show that smart meter data provides useful features in identification of consumer socio-demographic characteristics. Neale et al. (2022) use virtual smart meter data in building characteristic prediction task. They conclude that real smart meter data is preferable in prediction task. On the other hand, Wang et al. (2022) propose a semi-supervised learning approach for household profile identification because accurately labeled data can be expensive to collect. Sridhar et al. (2023) show that smart meter data can be used to classify the primary heating type of residential customers with an accuracy ranging from 75% to 88%. In general, more work is needed in quantifying the performance and suitability of supervised learning methods in customer identification based on electricity consumption time series.

Detailed and frequently measured consumption data bring new opportunities to utilize individual households also as a source for flexibility. As is noted in Biglaiser et al. (2018), data about consumer behavior make it easier to provide better services to the specific consumer groups. Incumbent firms with access to smart grid data, such as electricity retailers, may thus have a competitive advantage in markets for flexibility services. This ‘incumbency advantage’ arising from access to smart grid data should be included in the analysis of competition of flexibility services in the electricity markets.

In this article, we analyze two approaches for household’s main heating system classification: (1) time series analyses utilizing hourly electricity consumption data, and (2) classification based on socio-demographic and house features (house area, construction year, residence, income, and education). Our main contribution is to measure and compare the performance of these two approaches within the context of Finland. Our primary finding is that better main heat classification performance is achieved with features extracted from hourly smart meter data. Time series classification algorithms achieve 0.62 precision and 0.64 recall on average over the cross-validation test sets. Classification based on house and socio-demographic features leads to 0.41 precision and 0.43 recall. Secondly, if only house and socio-demographic features are available for the classification task, we show that the feature importance of construction year and house area is higher than the feature importance of income, education, and residence in city, population center, or dispersed settlement. Finally, given that we quantified the classification performance improvement capacity of hourly consumption time series we discuss the implications of this finding with respect to the competition policy.

2. Data and classification methods

To access electricity consumption as well as house and household specific data, we cooperated with a large distribution system operator Caruna Ltd. We approached with an online survey 10 679 of their residential customers in May 2018. The survey was sent to all Caruna Ltd customers who had already installed a grid-connected solar power system and whose email addresses were available (a total of 1 328 customers) and to a random sample of 9 342 their customers without a solar power system and whose email addresses were available. We received a response from 1 554 households and 1 361 of them gave permission to use their hourly electricity consumption data for research purposes. The electricity consumption data covers the period between 1.1.2014 and 31.12.2017. We use year 2017 consumption data in the time series classification task because utilizing the most recent consumption data assures that the collected background information from May 2018 does not deviate significantly from the respondent’s situation in year 2017.

For the analysis, we excluded households with a solar power system installed during the observation year 2017, households living in terraced houses or holiday homes, houses with living area above 400 m^2 or below 50 m^2, houses with an annual consumption below 1000 kWh, and households not responding to all background information questions. By removing these specific observations, we aim to ensure that the results are more robust in analyzing the effectiveness of house and socio-demographic features and electricity consumption time series in main heating mode classification. The final dataset includes 978 detached houses with house and socio-demographic background information, along with hourly electricity consumption data.

Main heating mode is used as the label in classification tasks. Table 1 presents the main heat distribution over the household sample. Direct electric heating, with a 41.2% share, is the most common heating mode. Heat pumps (ground heat, air heat, or exhaust air heat pump) and non-electric (district heating, oil, or wood) main heating systems are installed in roughly 50% of the households. Storage electric heating presents only 6.3% of the main heating labels in the sample.

Distributions of socio-demographic characteristics are presented in Table 2. The majority of houses in our research sample are in city area with more than 50,000 residents (Table 2, Residence). This may somewhat affect the heating system choice, because in these areas district heating is more often available than in the smallest population centers. However, it is more common to have wood-based heating or fireplaces as an auxiliary heating source in more sparsely populated areas.

Most households have a monthly gross income ranging from 4 000 € to 5 999 €, with only around 7% belonging to the highest group. The relatively high education level among Finnish citizens is seen also from our data as only 8.6% respondents belong to the lowest education group with the elementary school education only (Table 2, Education level). Over half of the respondents in our data have tertiary level education.

2.1. House and socio-demographic features

The floor area distribution conditional on the main heating is shown in Fig. 1. All main heating modes are utilized across houses with a considerable range of floor areas, but visual examination reveals certain variation, particularly between non-electric and heat pump houses. Houses with heat pump as the main heating mode have the largest average area with 175 m^2. Houses with non-electric heating or direct electric heating modes have the smallest average area of 147 m^2. We perform Mann-Whitney U and Kolmogorov-Smirnov tests to compare floor area distributions between non-electric and heat pump houses. The Mann-Whitney U test yields a p-value of 0.016, and the Kolmogorov-Smirnov test gives a p-value of 0.034. Both tests reject the null hypothesis, indicating a significant difference in distributions at a 5% significance level.

Construction year distribution conditional on main heating system is shown in Fig. 2. On average, houses with non-electric main heating mode are the oldest, with an average construction year of 1967. Conversely, houses with electric storage heating are the newest, with an average construction year of 1986.

Regardless of the residence, direct electric heating is the most

Table 1
Main heating mode distribution.

Main heating system	Description	Share (%)
Direct electric heating	Heating mode is direct electric heating	41.2
Storage electric heating	Heating mode is storage electric heating	6.3
Heat pump	Heating mode is ground, air or exhaust air heat pump	24.1
Non-electric	Heating mode is district heat, oil or wood	28.4

Table 2
Socio-demographic distribution.

Residence	Description	Share (%)
City	City, over 50 000 residents	36.7
DS	Dispersed settlement	29.2
City (small)	City, less than 50 000 residents	16.6
PC	Population center	12.6
PC (small)	Population center, less than 500 residents	4.9
Household gross income (€)	**Description**	**Share (%)**
< 4000	Monthly income below 4 000 €	23.9
< 6000	Monthly income between 4 000 and 5 999 €	32.6
< 8000	Monthly income between 6 000 and 7 999 €.	21.3
< 10,000	Monthly income between 8 000 and 9 999 €.	15.0
≥ 10,000	Monthly income above 10 000 €.	7.2
Education level	**Description**	**Share (%)**
1	Elementary school	8.6
2	Undergraduate / Vocational	27.0
3	Polytechnic	35.3
4	University	28.5
Other		0.6

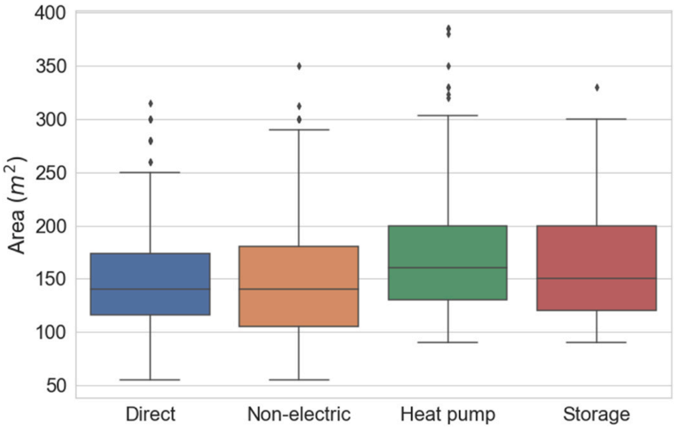

Fig. 1. House area distribution conditional on main heating mode. (Boxplot shows the 25th percentile, median and the 75th percentile of the dataset. Whiskers extend to points that lie within 1.5 interquartile ranges of the lower and upper quartiles).

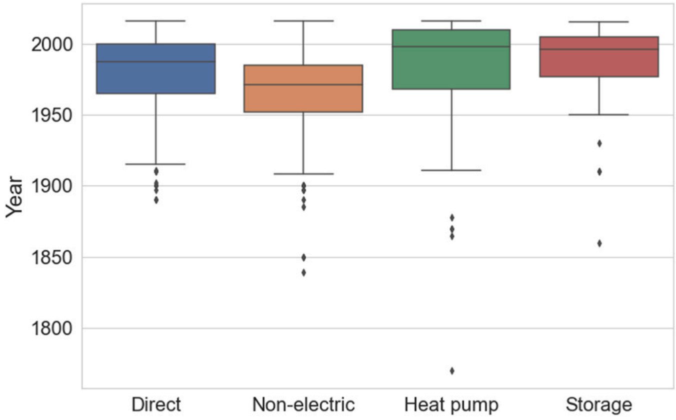

Fig. 2. House construction year distribution conditional on main heat. (Boxplot shows the 25th percentile, median and the 75th percentile of the dataset. Whiskers extend to points that lie within 1.5 interquartile ranges of the lower and upper quartiles).

popular heating system (Fig. 3). Non-electric main heat is the second most popular heat mode in cities and dispersed settlements, whereas in small cities and population centers heat pumps form the second most popular alternative.

Main heating shares according to household's monthly income is shown in Fig. 4. Direct electric heating has above 40% share among households with monthly income below 10 000€. The share of direct electric heating is only 33% in the top income group (>= 10 000 €). Heat pumps, with roughly 30% share, are more popular among households belonging to the two wealthiest income groups.

Main heating shares according to respondents' education level are shown in Fig. 5. Households with elementary school education (Education level 1) have non-electric heating as the largest class. Direct electric heating is clearly the most common heating choice among households with polytechnic degree (Education level 3).

Visual examination of the main heat distributions based on house and socio-demographic features indicates that these background factors can contribute to some extent identifying main heating labels. The distribution of main heating modes is displayed separately, and correlations should exist among the house and socio-demographic features. To assess the actual performance of house and socio-demographic factors as features in supervised learning classification algorithms, they are tested, and the classification results are presented in Section 3.1.

2.2. *Electricity consumption data*

The average annual electricity consumption is 14 113 kWh with standard deviation of 7 231 kWh in our data. Even though it is possible to choose several types of electricity contract, most households have a fixed price electricity contract (Table 3). The second most common contract is the Time-of-Use (Night) alternative, which offers lower prices during the night hours. The preference for this contract type may be explained by the fact that it has been available for a long time. Newer spot price contracts, in which prices fluctuate hourly based on the Nord Pool day-ahead price, were selected by 18.8% of the households. The least popular is the seasonal electricity contract, wherein the price varies depending on the season of the year.

The shape of the annual consumption distribution varies depending on the main heating system (Fig. 6). It is quite intuitive that the consumption is lowest among the non-electric heating households, as a significant portion of residential energy consumption is allocated to heating the premises. The electricity consumption is highest among households with storage electric heating. Typically houses belonging to this group are equipped with water circulating radiators and water is heated in large water tanks. On average, the annual consumption is 21 217 kWh for houses with storage electric heating, 8 149 kWh for non-electric heating houses, 16 575 kWh for houses with direct electric heating, and 14 373 kWh for houses with heat pumps.

Diverse main heating modes respond differently to outside temperature changes. To emphasize these potential variations in household electricity consumption, we calculate the standard deviation of hourly electricity use among houses with different heating systems. The standard deviation distributions for main heat classes are visualized in Fig. 7. Storage electric heating houses have the highest load variation, with an average standard deviation of hourly consumption at 2.31 kWh. In contrast, houses using non-electric main heating have a much lower consumption variation of only 0.77 kWh. These differences in electricity consumption variation underscore the impact of differing main heating modes on electricity consumption patterns.

Based on the distributions illustrated in Figs. 6 and 7, it seems evident that both the total and standard deviation of hourly consumption features could serve in identifying household main heating modes to some degree. However, there exists computationally efficient classification algorithms, using random sampling of features (Deng et al., 2013) or convolutional kernels (Dempster et al., 2019) extracted from time series. In Section 3.2, we present the main heat classification

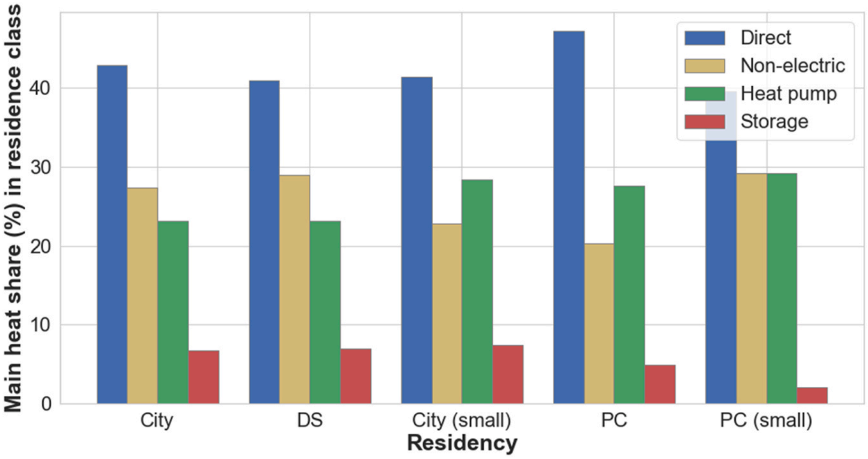

Fig. 3. Share of main heat in residence classes City, Dispersed settlement, City (small), Population center and Population center (small).

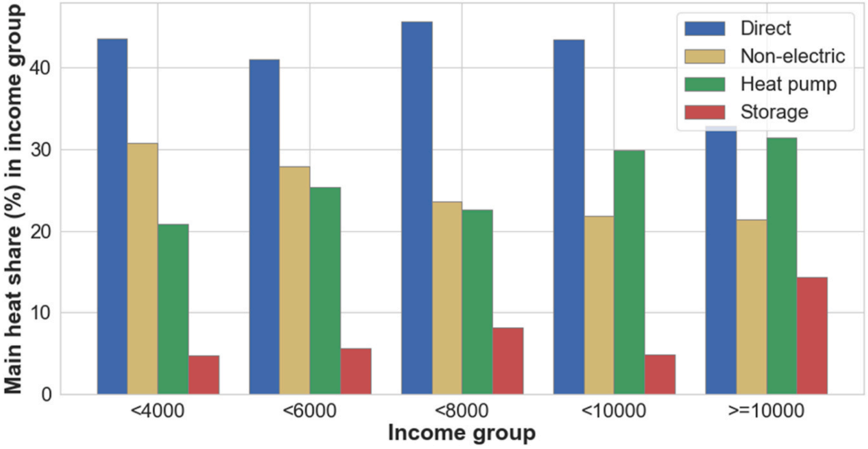

Fig. 4. Share of main heat under income groups (monthly income in euros).

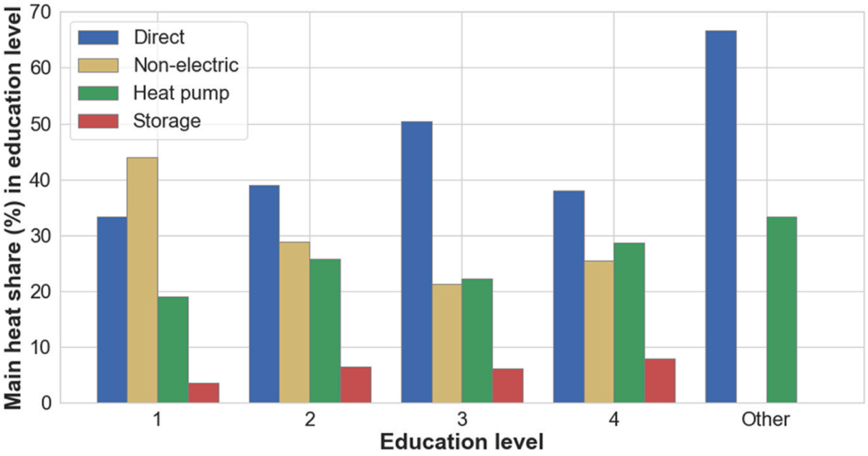

Fig. 5. Share of main heat under education levels 1 (elementary school), 2 (undergraduate/vocational), 3 (polytechnic), 4 (university) and Other.

performance of two time series classification methods, Time series forest and ROCKET, both applied to the hourly consumption dataset. The computational complexity for both algorithms is linear in time series length (Deng et al., 2013; Dempster et al.,2019).

2.3. *Classification methods*

We test three different types of classifier algorithms for the main heat identification task based on house and socio-demographic features: logistic regression, random forest, and neural nets (Geron, 2019). Scikit-learn machine learning library is used for classifier algorithm implementation (Pedregosa et al., 2011). Logistic regression classifier

Table 3
Household electricity contract distribution.

Electricity contract type	Percent
Fixed	44.0
Night	22.0
Spot	18.8
Seasonal	10.0
Do not know	5.2

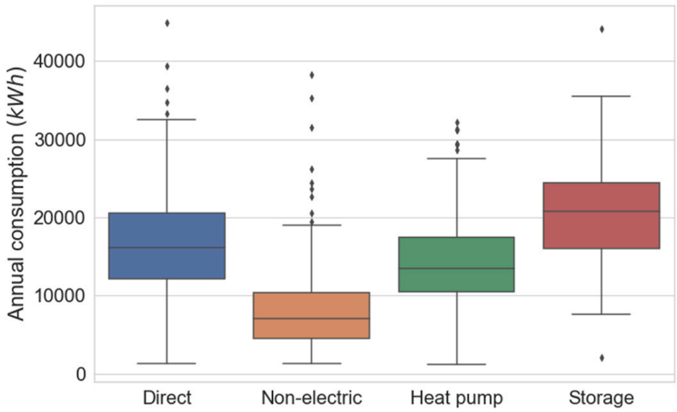

Fig. 6. Annual electricity consumption distribution conditional on main heat. (Boxplot shows the 25th percentile, median and the 75th percentile of the dataset. Whiskers extend to points that lie within 1.5 interquartile ranges of the lower and upper quartiles).

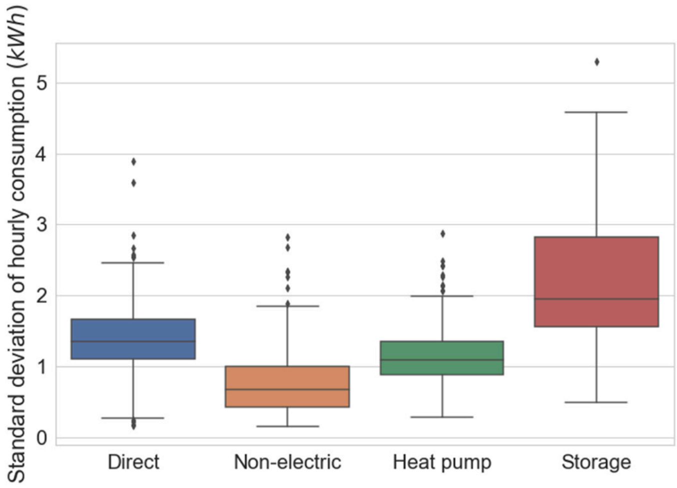

Fig. 7. Standard deviation of hourly electricity consumption distribution conditional on main heat. (Boxplot shows the 25th percentile, median and the 75th percentile of the dataset. Whiskers extend to points that lie within 1.5 interquartile ranges of the lower and upper quartiles).

minimizes the cross-entropy function, random forest classifier uses a random subset of features at each potential split and minimizes the gini impurity, and neural nets classifier minimizes the log-loss function using stochastic gradient descend. Features area and construction year are standardized by removing the mean and scaled to unit variance. Categorical features are processed by one-hot encoding.

We employ two computationally efficient time series classification methods in main heat classification based on hourly electricity consumption data. The first time-series method, Time series forest (TSF), randomly samples time series interval features mean, standard deviation and slope at each node of the classification tree. TSF method then utilizes entropy gain and distance measure for split selection (Deng et al., 2013). The second time-series method, ROCKET-classifier, transforms time series to random convolutional kernels (Dempster et al., 2020). Transformed features are then used to train a linear classifier with regularization. The computational complexity of TSF and ROCKET classifiers is linear in time series length, so they are scalable to long time-series.

In the classification task, we divide the entire dataset into training and test sets. The training dataset consists of 2/3 of the data, and the remaining 1/3 is designated for the test dataset. We implement a three-fold cross-validation approach to assess how well the model generalizes to different subsets of data. The analysis proceeds through three steps, with each fold used as a test set while the remaining two sets serve as the training set. We present the results as the mean values computed from the performance metrics evaluated on the test sets. For performance metric transparency, we do not optimize the hyperparameters, i.e., the default parameter values in sktime classes ComposableTimeSeriesForestClassifier and Rocket are used in all tested classifier algorithms (Löning et al., 2019).

3. Results and discussion

Our multi-label classification task is to predict the main heating class (direct, non-electric, storage, heat pump) based on the features available to the utilized classification algorithm. The performance of classification methods based on house and socio-demographic features is presented in Section 3.1. The performance metrics of time series classification methods and the confusion matrix are presented in Section 3.2.

We report performance metrics accuracy, precision, and recall for classification task evaluation. Accuracy gives the overall measure of how well the model can predict the correct class. Notice that a simple classifier which classifies all observations as the largest class in the sample, direct electric heating, would achieve an average accuracy of 41.2% (see Table 1).

Precision can be seen as a measure of quality and recall as a measure of quantity. In our research context precision metric informs which fraction of the specific heating system pointed out through analyses really belongs to that group. Recall metric informs the fraction of true main heat classes that were retrieved. The number of 1.0 would mean perfect precision, e.g., that every result retrieved by a search was relevant whereas a recall number 1.0 means that all relevant results were retrieved by the search. Because these metrics somewhat complement each other, their combined use in classification is recommended.

3.1. Classification based on house and socio-demographic features

Performance metrics of the main heat classification based on house and socio-demographic features are presented in Table 4. The model performance is presented as the mean value across the results of the three-fold cross-validation test set classifications. Direct electric heating can be classified with the highest precision and recall. Storage electric heating is systematically misclassified under other heating modes. The

Table 4
Results: main heat classification based on house and socio-demographic features.

Main heat class	Logistic regression		Random forest		Neural network	
	Precision	Recall	Precision	Recall	Precision	Recall
Direct	0.44	0.70	0.50	0.60	0.49	0.64
Heat pump	0.26	0.17	0.44	0.40	0.37	0.32
Non-electric	0.35	0.22	0.32	0.31	0.35	0.29
Storage	0.00	0.00	0.00	0.00	0.06	0.02
Macro average	0.26	0.27	0.31	0.32	0.32	0.32
Weighted average	0.34	0.39	0.41	0.43	0.40	0.42
Accuracy	0.39		0.43		0.42	

classification results underscore the impact of unbalanced sample data in relation to labels. Direct electric heating is the most frequent main heat class, whereas storage electric heating is the least frequent class. Random forest and neural net classifiers outperform logistic regression in the classification task. Among them, random forest exhibits the highest overall performance, achieving a weighted average precision of 0.41 and a recall of 0.43.

We assess the impact of the features on prediction accuracy by calculating the permutation importance of the random forest classifier. Permutation importance calculation is based on the fitted model. Each feature column in the test set is randomly shuffled with 30 repeats, and the decrease in model performance is measured to determine the permutation importance of each feature (Pedregosa et al., 2011). Only construction year and living area have positive permutation importance within the 95% confidence interval. The average permutation importance is 0.07 for construction year and 0.04 for living area. When compared to the 0.43 accuracy achieved by the random forest classifier (see Table 4), it can be concluded that both these features offer distinct information for the main heat classification task. The overall correctness of the classification model decreases to 0.36 if construction year is not available and to 0.39 if living area is not included as a feature. However, given that the drop in the model accuracy is not substantial suggests that house and socio-demographic features are correlated with each other. We present the Spearman correlation between these features in Appendix A.

3.2. Hourly consumption time series

Precision and recall performance metrics of main heat classification based on features extracted from hourly consumption time series are presented in Table 5. Both time series classification methods clearly outperform the classification methods based on house and socio-demographic features. ROCKET method slightly outperforms TSF method in terms of classification accuracy. ROCKET method achieves a weighted average precision of 0.62 and a recall of 0.64. This represents a significant improvement compared to the random forest classifier's performance based on house and socio-demographic features, which achieved a precision of 0.41 precision and a recall of 0.43 (Table 4). Precisions values ranging from 0.60 to 0.68 are achieved for the classification of direct electric, heat pump and non-electric heat modes, while recall values vary between 0.52 to 0.84.

The confusion matrix is a visual representation that displays the distribution of cases that are incorrectly classified. Correctly classified cases appear along the diagonal of the confusion matrix. Any off-diagonal cases indicate misclassifications. Precision measures the proportion of correctly classified cases in relation to the predicted classes in the columns, while recall measures the proportion of correctly classified cases relative to the actual classes in the rows. Fig. 8 shows a convolutional kernel (ROCKET) classifier confusion matrix for one of the folds. Notably, a significant number of non-electric main heat houses are misclassified as direct electric heating houses, and heat pumps are misclassified both as direct heat and non-electric heat houses.

Table 5 shows that ROCKET method classifies storage electric heat, the smallest class in the data sample (constituting 6.3% of houses), with low precision (0.11) and recall (0.05). In contrast, TSF method achieves a notably better performance in storage heat classification with 0.63 precision and 0.24 recall. Consequently, when one uses the default hyperparameter settings in the sktime framework (see Löning et al., 2019), TSF times series classification method appears to be more favorable than ROCKET method, when achieving a high classification performance for the smallest main heat class is of high importance.

Table 5
Results: main heat classification based on hourly consumption data.

Main heat class	Random convolutional kernels (ROCKET)		Time series forest (TSF)	
	Precision	Recall	Precision	Recall
Direct	0.65	0.84	0.58	0.74
Heat pump	0.68	0.61	0.72	0.65
Non-electric	0.60	0.52	0.43	0.38
Storage	0.11	0.05	0.63	0.24
Macro average	0.51	0.51	0.59	0.50
Weighted average	0.62	0.64	0.60	0.59
Accuracy	0.64		0.59	

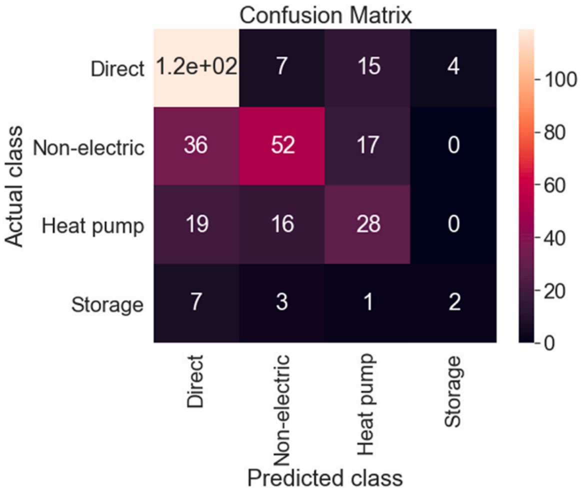

Fig. 8. Random convolutional kernels classifier confusion matrix.

We perform a sensitivity analysis regarding the length of the consumption time series data. The accuracy of the TSF classifier improves from 0.59 to 0.62, when data from 2014 to 2017 is utilized (Appendix A., Table A2). A slight improvement in performance might encourage the use of longer time series in main heat classification. However, one should consider trends in household investments, especially during a green transition, where investments in efficiency and electrification of heating may influence the electricity consumption pattern over time.

3.3. Comparison of the main heat classification methods

The applicability of the tested classification algorithms is contingent

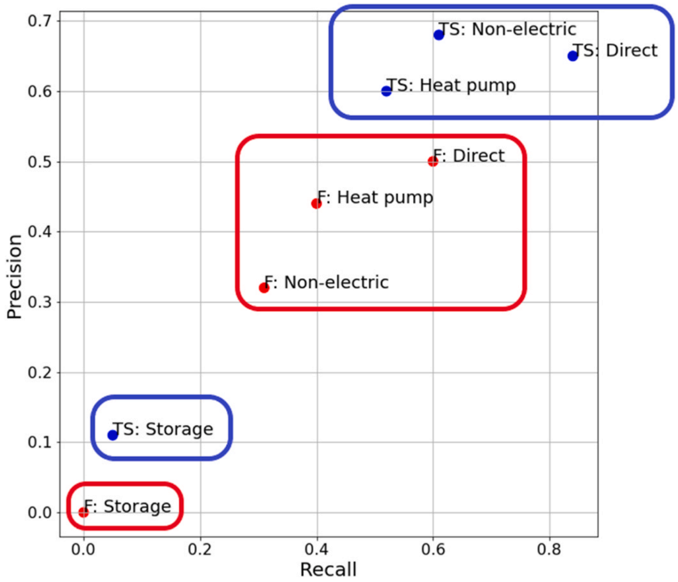

Fig. 9. Precision and recall for main heat classes with classification based on house and socio-demographic features with random forest classifier (F, red) and electricity consumption time series with ROCET classifier (TS, blue).

upon the specific use case. Fig. 9 shows that classification algorithm using house and socio-demographic features (F, red) can achieve a 0.50 precision rate and a 0.60 recall rate in direct electric heating classification task. For some applications, this performance level may be sufficient. However, time series classification (TS, blue) clearly outperforms main heating classification method based on house and socio-demographic features, as all three largest main heating classes, i. e., direct, heat pump and non-electric heating, can be classified with ≥ 60 precision rate and ≥ 52 recall rate. This result indicates that consumption time series data can provide value in targeting potential households for flexibility services.

In Finland, smart meter data is stored in a centralized information exchange system known as Datahub (Fingrid Datahub Services, 2022). Electricity customers can grant authorization to potential service providers to access their customer data stored in Datahub, enabling the provision of various services to the customer. Based on our classification results, the possibility for data sharing seems well founded, as service providers can better screen potential customers. Thus, a better match between flexibility services and flexibility resources can be achieved through consumption data sharing.

4. Conclusions

Value of flexibility increases with higher share of renewable intermittent energy sources in the electricity system. Generally, home heating provides a great and underutilized source of demand-side flexibility in the residential sector. So far, the literature has lacked analysis testing the performance of different supervised learning algorithms to identify the residential flexibility potential rising from home heating systems. Hence, this paper examines varying supervised learning algorithms in the main heating classification task. We use a detailed dataset of Finnish detached houses and test two sources of classification algorithm input data: i) socio-demographic and house characteristics background features, and ii) features extracted from the consumption time series. Results show that a better main heat classification performance, both in precision and recall, can be achieved with the time series classification methods. In the future, it is important to test the applicability and generalizability of the main heat classification algorithms across different countries with diverse heating demands and building types.

The classification results demonstrate that hourly consumption data can serve as valuable input for identifying houses suitable as flexibility resources. In consequence, data availability can be a concern from the competition policy point of view. If only incumbent retailers have access to electricity consumption time series data, entry of potential flexibility service providers may be hindered due to reduced information base for household targeting. We emphasize the importance of granting customers the authority to approve the sharing of consumption time series data in centralized electricity market information exchange systems.

The practical usability of tested classification methods depends on the application at hand. Time series classification ROCKET performed with over 0.60 precision rate and 0.52 recall rate for direct, non-electric and heat pump houses. However, storage electric heat houses, which represented the smallest class in the sample, are classified with poor performance. On the other hand, time series forest classifier achieves a better classification performance with respect to the smallest storage heat class.

In this study we utilized the default parameter setting in the classification tasks. Regarding classification based on house and socio-demographic features (Section 3.1), it is likely that the three tested algorithms captured most of the signal within the data. In the context of time series classification (Section 3.2), hyperparameter optimization could improve the performance, considering the various ways to extract features from time series data. For the TSF algorithm, increasing the number of trees in the forest could improve performance, albeit with an associated increase in computational cost. For the ROCKET algorithm, increasing the number of kernels could enhance classification performance. Again, this results in an increase in computational cost. An interesting future study is to optimize the hyperparameters of the time series classification algorithms so that classification performance of the minor classes does not drop when imbalanced main heat data is used.

Finally, we focused on the latest available data from 2017 for our time series classification analysis to align with household background information collected in May 2018. Testing the TSF classifier with consumption data spanning 2014–2017 (see Appendix A, Table A2) revealed an improvement in accuracy from 0.59 (2017 data alone) to 0.62 (2014–2017 data). This suggests that longer time series positively impact the classifier's performance in our study. However, in dynamic environments like electricity markets, the influence of additional time series data on classification performance may not consistently follow a positive trajectory. A promising direction for research is investigating how classification performance changes by incorporating longer time series data in electricity system applications.

CRediT authorship contribution statement

Samuli Honkapuro: Writing – original draft, Project administration, Funding acquisition, Conceptualization. **Hannu Huuki:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Maria Kopsakangas-Savolainen:** Writing – original draft, Project administration, Funding acquisition, Data curation, Conceptualization. **Enni Ruokamo:** Writing – original draft, Project administration, Data curation, Conceptualization. **Araavind Sridhar:** Writing – original draft, Conceptualization. **Nadezda Belonogova:** Writing – original draft, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to proofread specific paragraphs. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work has been funded by Business Finland in project HOPE (funding decision 22693/31/2020) and by Academy of Finland in project EasyDR (348100).

Appendix A

Table A1
Spearman rank correlation.

	Area	Year	Education	Income
Area	1			
Year	0.277	1		
Education	0.251	0.138	1	
Income	0.390	0.283	0.404	1

Table A2
Time series forest classification result: 2017 vs. all (2014–2017) data.

Main heat class	Time series forest (TSF) 2017 consumption data		Time series forest (TSF) 2014-2017 consumption data	
	Precision	Recall	Precision	Recall
Direct	0.58	0.74	0.62	0.77
Heat pump	0.72	0.65	0.72	0.70
Non-electric	0.43	0.38	0.52	0.44
Storage	0.63	0.24	0.58	0.27
Macro average	0.59	0.50	0.62	0.54
Weighted average	0.60	0.59	0.65	0.62
Accuracy	0.59		0.62	

References

ACER, 2022, Annual Report on the Results of Monitoring the Internal Electricity and Natural Gas Markets in 2021, Energy Retail and Consumer Protection Volume. Available: ⟨https://www.acer.europa.eu/sites/default/files/documents/Publications/MMR_2021_Energy_Retail_Consumer_Protection_Volume.pdf⟩.

Biglaiser, G., Calvano, E., Cremer, J., 2018. Incumbency advantage and its value. J. Econ. Manag. Strategy 28, 41–48.

Carrol, P., Murphy, T., Hanley, M., Dempsey, D., Dunne, J., 2018. Household classification using smart meter data. J. Off. Stat. 34, 1–25.

Dempster, A., Petitjean, F., Webb, G.I., 2020. ROCKET: exceptionally fast and accurate time series classification using random convolutional kernels. Data Min. Knowl. Discov. 34, 1454–1495.

Deng, H., Runger, G., Tuv, E., Vladimir, M., 2013. A time series forest classifier and feature extraction. Inf. Sci. 293, 142–153.

Fingrid, 2022, Fingrid Electricity System Scenarios Draft, ⟨https://www.fingrid.fi/en/news/news/2022/future-scenarios-for-finlands-electricity-system-published/⟩.

Fingrid Datahub Services, 2022, Datahub connects energy providers, ⟨https://palvelut.datahub.fi/en/⟩ [accessed 3 January 2023].

Finnish Energy, 2022 Energy Year 2021 – Electricity, ⟨https://energia.fi/en/newsroom/publications/energy_year_2021_-_electricity.html#material-view⟩.

Geron, A., 2019. Hands-on Machine Learning with Scicit-learn, Keras & Tensorflow. Concepts, Tools, and Techniques to Build Intelligent Systems, 2nd ed. O'Reilly.

Huuki, H., Karhinen, S., Kopsakangas-Savolainen, M., Svento, R., 2020. Flexible demand and supply as enablers of variable energy integration. J. Clean. Prod. 258, 120574.

IEA (International Energy Agency), 2022a, Heating - Fuels & Technologies - IEA, Accessed 23.1.2023.

IEA (International Energy Agency), 2022b, World Energy Outlook 2022, ⟨https://iea.blob.core.windows.net/assets/830fe099–5530-48f2-a7c1–11f35d510983/WorldEnergyOutlook2022.pdf⟩.

Känsälä, K., Hammar, K., 2018, Demand Management for Future Electricity Market. VTT Technical Research Centre of Finland. VTT White Paper. Demand Management for Future Electricity Market — VTT's Research Information Portal.

Löning, M., Bagnall, A., Ganesh, S., Kazakov, V., Lines, J., Kiraly, F., 2019, sktime: A Unified Interface for Machine Learning with Time Series. 33rd Conference on Neural Information Processing Systems (NeurIPS 2019), Vancouver, Canada.

Mahapatra, K., Gustavsson, L., 2008. An adopter-centric approach to analyze the diffusion patterns of innovative residential heating systems in Sweden. Energy Policy 36 (2), 577–590. https://doi.org/10.1016/j.enpol.2007.10.006.

Meles, T.H., Ryan, L., Mukherjee, S.C., 2022. Heterogeneity in preferences for renewable home heating systems among Irish households. Appl. Energy 307, 118219. https://doi.org/10.1016/j.apenergy.2021.118219.

Michelsen, C.C., Madlener, R., 2012. Homeowners' preferences for adopting innovative residential heating systems: a discrete choice analysis for Germany. Energy Econ. 34 (5), 1271–1283. https://doi.org/10.1016/j.eneco.2012.06.009.

Neale, A., Kummert, M., Bernier, M., 2022. Discriminant analysis classification of residential electricity smart meter data. Energy Build. 258, 111823.

Ortega-Izquierdo, M., Paredes-Salvador, A., Montoya-Rasero, C., 2019. Analysis of the decision making factors for heating and cooling systems in Spanish households. Renew. Sustain. Energy Rev. 100, 175–185. https://doi.org/10.1016/j.rser.2018.10.013.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, E., 2011. Scikit-learn: machine learning in Python. J. Mach. Learn. Res. 12, 2825–2830.

Rosenow, J., Gibb, D., Nowak, T., Lowes, R., 2022. Heating up the global heat pump market. Nat. Energy 7, 901–904.

Ruokamo, E., 2016. Household preferences of hybrid home heating systems – a choice experiment application. Energy Policy 95, 224–237.

Räihä, J., Ruokamo, E., 2021. Determinants of supplementary heating system choices and adoption consideration in Finland. Energy Build. 251, 111366.

Sahari, A., 2019. Electricity prices and consumers' long-term technology choices: evidence from heating investments. Eur. Econ. Rev. 114, 19–53. https://doi.org/10.1016/j.euroecorev.2019.02.002.

Sridhar, A., Belonogova, N., Honkapuro, S., Huuki, H., Kopsakangas-Savolainen, M., Ruokamo, E., 2023. Identifying hybrid heating systems in the residential sector from smart meter data. J. Build. Eng. 74, 106867.

Statistics Finland, 2022, Energy Consumption in Households, PxWeb - Select table (stat.fi). Accessed 23.1.2023.

Tureczek, M., Nielsen, P.S., 2017. Structured literature review of electricity consumption classification using smart meter data. Energies 10, 584.

Wang, Y., Chen, Q., Gan, D., Yang, J., Kirschen, D.S., Kang, C., 2019. Deep learning-based socio-demographic information identification from smart meter data. IEEE Trans. smart grid Vol. 10.

Wang, F., Lu, X., Chang, X., Cao, X., Yan, S., Li, K., Duic, N., Shafie-khah, M., Catalao, P. S., 2022. Household profile identification for behavioral demand response: a semi-supervised learning approach using smart meter data. Energy 238 (Part B), 121728.

Hannu Huuki is a Ph.D. candidate at the University of Oulu Business School, Finland. He works as a researcher at the Finnish Environment Institute. His research interests are electricity markets, demand side management and dynamic optimization of flexible energy resources.

Enni Ruokamo holds a Ph.D. in Economics from University of Oulu with specialization in energy and environmental economics. She works as a Senior Research Scientist at the Finnish Environment Institute Syke. Her research focuses on household energy use aspects, public preferences for environmental goods and empirical studies on environmental policies.

Maria Kopsakangas-Savolainen is a Professor of Energy Economics at the Finnish Environment Institute and the University of Oulu Business School in Finland. Recently main emphasis of her research has concerned on the promotion of flexibility in the energy system, on consumers preferences towards energy contracts, value of intermittent energy, promotion market access for renewable energy and efficient use of renewable energy. She has large teaching and professional experience and is Associate Editor of the journal Energy Efficiency (Springer) and member of the board in the journal Finnish Economic Papers.

Nadezda Belonogova has been working as a junior researcher and a post-doctoral researcher in LUT University, Laboratory of Electricity Markets and Power Systems, in 2009–2018 and 2019–2023, respectively. Her research interests are smart grids, electricity markets and use of machine learning algorithms to estimate flexibility domain of residential consumption.

Samuli Honkapuro is a tenured professor in Energy Markets and Energy Systems in LUT University, School of Energy Systems, Lappeenranta, Finland. In addition, he is the head of the Laboratory of Electricity Markets and Power Systems in LUT. He graduated as D.Sc. (tech.) from LUT in 2008, and since then, he has been carrying out scientific research and academic education. His key research interests are electricity market design and business models for integration of distributed energy resources, demand side management, and energy communities.

Araavind Sridhar is a Ph.D. candidate between LUT University, Finland and Politecnico di Milano, Italy. He received his masters degree in sustainable energy engineering from KTH Royal Institute of Technology, Sweden. His research interests are demand side management within the residential sector, electricity markets, optimization of distributed energy resources and machine learning.